```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
from google.colab import files
import zipfile
import io

uploaded = files.upload()

# Assuming only one zip file is uploaded and it contains 'insurance.csv'
# Get the filename of the uploaded zip file
zip_file_name = next(iter(uploaded))

# Unzip the file
with zipfile.ZipFile(io.BytesIO(uploaded[zip_file_name]), 'r') as z:
    z.extractall()

df = pd.read_csv("insurance.csv")
df.head()
```

Choose Files archive (1).zip

archive (1).zip(application/x-zip-compressed) - 16425 bytes, last modified: 3/17/2026 - 100% done

```
Saving archive (1).zip to archive (1) (2).zip
```

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

Next steps: Generate code with df New interactive sheet

```
# Convert categorical variables into numeric
df_encoded = pd.get_dummies(df, columns=['sex','smoker','region'], drop_first=True)

df_encoded.head()
```

	age	bmi	children	charges	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.900	0	16884.92400	False	True	False	False	True
1	18	33.770	1	1725.55230	True	False	False	True	False
2	28	33.000	3	4449.46200	True	False	False	True	False
3	33	22.705	0	21984.47061	True	False	True	False	False
4	32	28.880	0	3866.85520	True	False	True	False	False

Next steps: Generate code with df_encoded New interactive sheet

```
X = df_encoded[['age','bmi','children','smoker_yes']]
y = df_encoded['charges']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

```
model = LinearRegression()
model.fit(X_train, y_train)
```

▾ LinearRegression i ?

LinearRegression()

```
y_pred = model.predict(X_test)
```

```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("MSE:", mse)
print("R2 Score:", r2)
```

```
MAE: 4213.798594527248
MSE: 33981653.95019776
R2 Score: 0.7811147722517886
```

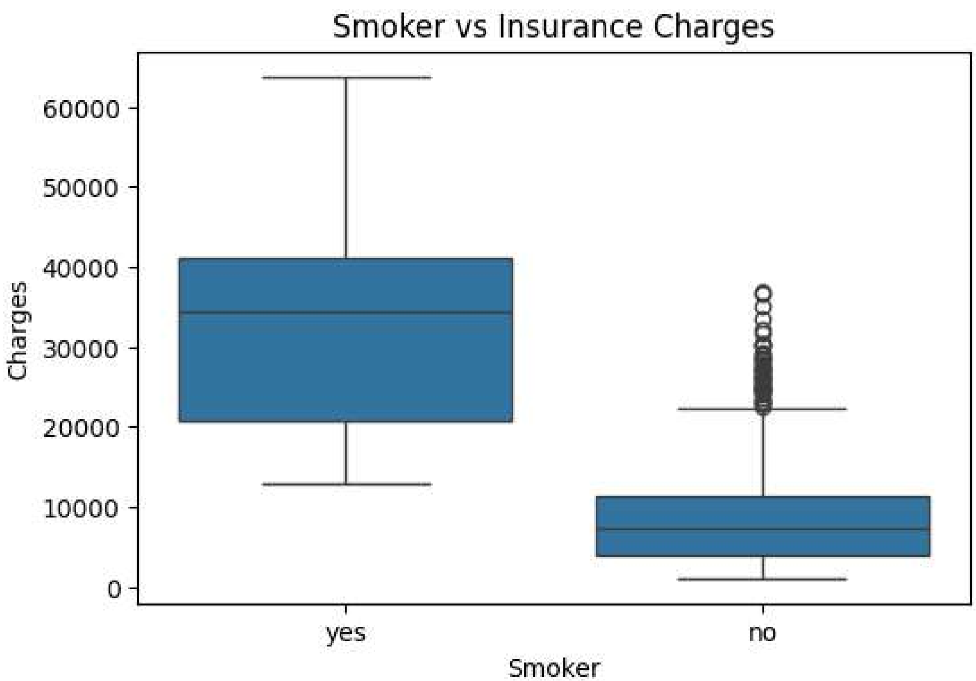
```
coefficients = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.coef_
})

print(coefficients)
```

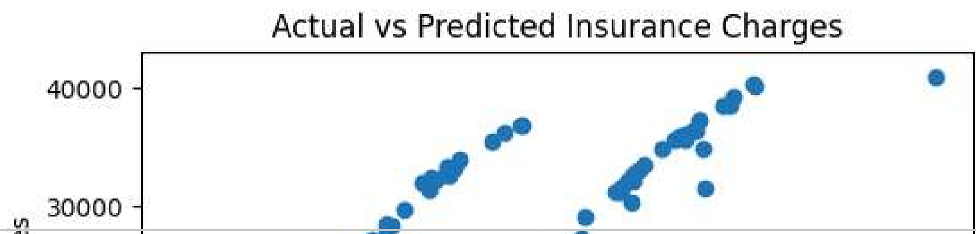
```
      Feature   Coefficient
0         age    257.071158
1         bmi    327.522631
2    children    427.199971
3  smoker_yes  23653.211646
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
plt.figure(figsize=(6,4))
sns.boxplot(x=df['smoker'], y=df['charges'])
plt.title("Smoker vs Insurance Charges")
plt.xlabel("Smoker")
plt.ylabel("Charges")
plt.show()
```




```
plt.figure(figsize=(6,4))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Charges")
plt.ylabel("Predicted Charges")
plt.title("Actual vs Predicted Insurance Charges")
plt.show()
```




```
plt.figure(figsize=(6,4))
sns.scatterplot(x=df['age'], y=df['charges'])
plt.title("Age vs Insurance Charges")
plt.xlabel("Age")
plt.ylabel("Insurance Charges")
plt.show()
```